Comprehensive Exam Preparation Guide: Social and Professional Issues in Computing (SPIC)

This guide provides a exhaustive, lecture-by-lecture breakdown of all study materials in your folder, followed by detailed, fully-solved answers to all four previous midterm exam papers.

Part 1: Exhaustive Lecture-by-Lecture Study Materials

Chapter 1: Benefits of Computer Technology

Week 1: Unwrapping the Gift

- **Ubiquity of Technology:** Computers are embedded in daily life. Examples include ATMs, CD players, mobile phones, modern home appliances, and music players (i.e., iPods).
- **Direct vs. Indirect Use:** Direct use involves actively interacting with a system (e.g., using a smartphone). Indirect use occurs when a third-party system utilizes computer technology on your behalf (e.g., credit card transactions processed by banks, or utilities managed by smart grids).
- **Negative Impacts of New Technology:**

1. *Unemployment:* Job displacement due to automation.

2. *Alienation:* Reduced face-to-face human interaction.

3. *Poor Customer Service:* Over-reliance on automated menus and chatbots.

4. *Crime:* Cyberattacks, digital scams, and identity theft.

5. *Loss of Privacy:* Constant tracking and digital records.

6. *Errors:* System bugs causing financial or operational disruptions.

- **General Positive Benefits:**

* Convenience and reduced manual effort.

* Significant time and cost savings.

* Creation of new job classes (e.g., software engineers, data analysts).

* Increased options for transactions (e.g., online banking, e-commerce).

* Improvements in crime-fighting.

* Improved customer service through immediate access to accounts.

* Fewer manual errors and improved entertainment options.

- **Appreciating the Benefits in Key Sectors:**

1. **Communication:** Offers non-invasive communication (read messages at convenience), time-saving connectivity, access to immense info, combined media (text/sound/graphics), free web info, borderless public forums, and remote access.

2. **Transportation:** Navigation (GPS), diagnostics (chips tracking engine behavior), safety (ABS - Anti-lock Braking Systems), hybrid engines, fuel efficiency, and traffic pattern studies.

3. **Education:** Language acquisition, literacy tools, distance learning, simulations (e.g., training air traffic controllers or endoscopic surgeons), and speech recognition/synthesis.

4. **Fighting Crime:** Improved crime reporting, faster searches of arrest and fingerprint databases, remote access from patrol cars, enhanced record-keeping (photos/fingerprints), and advanced surveillance equipment.

5. **Medicine:** Sophisticated imaging (MRI, CT scans), reduced surgery time, faster recovery (minimally invasive surgery), patient monitoring, improved treatments, robotic pharmacies, digital patient records (legible, organized, accessible, immediate), streamlined diagnosis, and telemedicine.

6. **Disabled Support:** Braille printers, speech control for appliances, mobility assistance, artificial limb control, large font displays, speech input/synthesis, and opportunities to return to the workplace.

7. **Automation:** Inventory organization, cost/time savings, tedious work handled by machines, and improved workplace safety.

8. **Identifying People/Products:** Bar codes & smart cards (Time-saving, accuracy, ease of use, cost-saving, secure, customizable, reliable).

9. **Environmental Monitoring:** Automated tracking of temperature, acceleration, stress in materials, emissions, air substances, moisture, and acidity.

10. **Other Benefits:** Tracking rare plants/animals, migration habits, stolen goods, and lost pets/people. Reduction in paper waste and toxic wastes from paper mills.

Chapter 2: Privacy

Week 2 Lecture 1: Privacy and Computer Technology

- **Three Key Aspects of Privacy (Sara Baase):**

1. *Freedom from intrusion:* Being left alone.

2. *Control of information about oneself:* Managing who gets access to your personal details.

3. *Freedom from surveillance:* Not being watched, tracked, or recorded.

- **Categories of Privacy Threats:**

1. *Intentional, institutional uses:* Authorized collection by governments or businesses.

2. *Unauthorized use or release by insiders:* Employees leaking or selling databases.

3. *Theft of information:* External hackers or physical device theft.

4. *Inadvertent leakage:* Accidental data exposures (e.g., misdirected emails).

5. *Our own actions:* Voluntarily posting personal information online.

- **New Technology, New Risks:**

* Government/private databases containing detailed personal profiles.

* Sophisticated surveillance tools: cameras, GPS, smartphones.

* Search query data (terabytes collected daily for target ads and services).

* Data stored and transmitted without users' knowledge or explicit consent.

- **Terminology of Data Collection:**

Invisible Information Gathering:* Collecting personal details without the user's knowledge (e.g., cookies, spyware, ISP logging).

Secondary Use:* Using data for a purpose other than the one it was provided for (e.g., selling medical history to insurance marketers).

Data Mining:* Searching and analyzing databases to find patterns and build knowledge.

Computer Matching:* Combining info from different databases using a unique identifier (like SSN or NID).

Computer Profiling:* Using database records to predict likely human behaviors or target specific demographics.

- **Consent Mechanisms:**

Informed Consent:* Users are notified of privacy policies and then choose whether to interact.

Opt-in Policy:* User must take affirmative action (check a box) to consent. Safe-by-default.

Opt-out Policy:* Data is collected unless the user takes action to stop it. Surveillance-by-default.

- **Fair Information Principles (FIPs):**

1. Inform users when data is collected, what is collected, and how it is used.

2. Collect only needed data.

3. Offer opt-out options for secondary uses and features that expose personal data.

4. Keep data only as long as needed (Data Retention).

5. Secure data from theft and accidental leaks, with stronger protection for sensitive data.

6. Establish policies for law enforcement requests.

Week 2 Lecture 2: "Big Brother is Watching You"

- **Government Databases:** Government agencies collect data, buy information from private sellers, and request business reports to conduct data mining and computer matching (e.g., fighting terrorism).
- **Consequences:** Government databases can lead to arrests, jailing, or asset seizures. This shifts the legal landscape from a presumption of innocence to a presumption of guilt. Computer profiling can subject innocent people to embarrassing searches and costly investigations.
- **Surveillance Examples:** Tax records, public library records, medical files, firearms permits, marriage/divorce records, loan applications, property ownership, professional licenses, welfare databases, school history, and voter registration files.
- **Surveillance Problems:** Personal data no longer resides safely in our homes; it lives in databases outside our control. New tech allows government agencies to search our homes and monitor us from a distance without our

knowledge.

- **Video Surveillance:** Malls, shopping centers, and banks use cameras. Combining cameras with facial recognition increases security but severely decreases personal privacy.

Week 3 Lecture 3: Diverse Privacy Topics

- **Dossiers & Targeted Marketing:** Data firms buy consumer data, build detailed dossiers, scan email content (e.g., Gmail ads), and sell profile data to marketers.
- **Social Networks:** Users publish personal info (gossip, pictures, "away from home" status), while platforms implement complex, unexpected privacy settings.
- **Location Tracking:** GPS, cell phones, and mobile apps track real-time locations, selling data to advertisers.
- **Stolen/Lost Data:** Arises from hacking, physical theft of laptops/drives, pretexting (phishing), and employee bribery.
- **What We Do Ourselves:** Users often lack awareness of cyberspace longevity. Young people put less value on privacy, posting pictures/blogs, and storing files in cloud systems outside their direct control.
- **Public Records:** Bankruptcy, property, and arrest records are online. While transparent, they present a major risk of identity theft.
- **Children's Privacy:** Minors are vulnerable to online predators and unable to make informed decisions. Parental monitoring (webcams, GPS tracking, RFID) raises ethical questions about when tracking violates a child's privacy and affects their development of independence.
- **National ID System:**

SSNs:* Too widely used and easy to falsify.

Pros of National ID:* Harder to forge, consolidates identification cards.

Cons of National ID:* Threat to general freedom/privacy, high potential for centralized abuse.

- **Technology & Markets:** Users can adopt privacy-enhancing technologies (PETs) like cookie disablers, pop-up blockers, and spyware scanners.
- **Encryption:** Hides data from interception using cryptography.

Public-key Cryptography:* Uses a public key for encryption and a private key for decryption.

- **Business Safeguards:** Unique user IDs, access control (limiting permissions based on roles, e.g., billing clerks don't need lab tests), audit trails (records of who accessed and modified data), and independent privacy audits.
- **Rights & Laws:**

Warren and Brandeis (The Inviolate Personality):* Arguing privacy is a distinct right. People should have the right to prohibit the publication of personal facts, appearance, and relationships.

Judith Jarvis Thomson:* Arguing there is no distinct right to privacy; a privacy violation is always a violation of another basic right (e.g., property rights, person, or contracts).

Regulation:* Laws like HIPAA (Health Insurance Portability and Accountability Act) in the US protect sensitive health records.

Wiretapping Laws:* 1934 Communications Act prohibited interception; 1968 Omnibus Crime Control Act allowed wiretapping with a court order; 1986 Electronic Communications Privacy Act (ECPA) protected email.

Contrasting Views:* Free Market View (voluntary agreements, consumer choices) vs. Consumer Protection View (strong regulation to combat data errors, leaks, and consumer lack of knowledge).

Chapter 3: Technology & Free Speech

Week 4 Lecture 1: Changing Paradigms & Censorship

- **The Traditional Three-Part Regulatory Framework:**

1. *Print Media:* Strongest constitutional protection; free from government content control.

2. *Broadcast Media:* Government licenses bandwidth and regulates structure and content (e.g., Canadian content laws, language restrictions).

3. *Common Carriers:* Transmit messages without controlling content (e.g., telephone system). The carrier is not responsible for user statements.

- **The Internet Dilemma:** BBS and the World Wide Web do not fit these paradigms, raising platform liability debates.
- **Canada Charter Section 2(b):** Guarantees freedom of thought, belief, opinion, and expression.
- **Motivations for Free Speech (Chief Justice McLachlan in *R. v. Keegstra*):**

1. Essential to political democracy and democratic institutions.

2. Promotes a marketplace of ideas to search for truth.

3. Key to individual self-actualization.

4. Prevents state suppression and hiding of mistakes.

- **Obscenity in Canada (s. 163(8)):** Defined as the undue exploitation of sex, or sex combined with crime, horror, cruelty, and violence.
- **Canadian Cases:**

Little Sisters Bookstore:* Customs targeted gay/lesbian imports, ruled systematically discriminatory.

R. v. Sharpe:* Written descriptions of morally repugnant acts are not illegal unless they advocate or counsel the acts.

- **Censorship Laws (US):**

Communications Decency Act (CDA, 1996):* Ruled unconstitutional (censored adult speech).

Child Online Protection Act (COPA, 1998):* Enjoined by courts.

Children's Internet Protection Act (CIPA, 2002):* Constitutional; schools/libraries must use filters to receive federal funding.

- **Canada CRTC (1999):** Chose *not* to regulate internet content, treating it as an open common carrier.
- **Filtering Software:** Blocks illegal or harmful content but suffers from over-blocking (stopping educational search queries) and under-blocking.
- **Hate Speech in Canada:**

Criminal Code s. 318 & 319:* Illegal to advocate genocide, publicly incite hatred, or promotion of hatred against an identifiable group (based on color, race, religion, ethnic origin). s. 320.1 allows removal of hate propaganda.

Canadian Human Rights Act Section 13:* Prohibits hate messages via telecommunications (e.g., Ernst Zundel case, though his site remained up in the US).

Week 4 Lecture 2: Anonymity, Spam, & Diversity

- **Anonymity Range:**

True Anonymity (Unlinkability):* No one can determine the author's identity. Legally problematic.

Pseudonymity:* Persistent identifier that is not the real name.

Public Pseudonym:* Link easily discoverable by the public.

Non-public Pseudonym:* Link known only to system administrators.

Unlinkable Pseudonym:* Link not recorded or discoverable by anyone.

Anonymous Re-mailers:* Strips headers and forwards emails (e.g., Johan Helsingius's

1993 Finnish service).

- **Anonymity Arguments:**

Pros:* Protects whistleblowers, human rights workers, abuse victims, and prevents discrimination.

Cons:* Shields harassers, enables fraud, and ruins civil discourse in the marketplace of ideas.

- **Spam (Unsolicited Mass Email):** Imposes time and account costs on users, plus bandwidth costs on ISPs.

AOL v. Cyber Promotions:* AOL won due to direct economic harm to their servers.

Intel Employee Case:* Sending non-commercial emails to Intel servers was not trespass because it caused no economic harm.

Solutions:* Filters, blocklists, CAN-SPAM Act (US, opt-out), PIPEDA (Canada, opt-in/consent).

- **Ensuring Diversity:** Debates on net neutrality (tiered service), public subsidies for arts/culture sites, and ensuring civic information sites exist.

Chapter 4: Computer Crime

Week 5 Lecture 1: Hacking history, US & Canada laws

- **Hacking Phases:**

Phase 1 (1960s-1970s):* Joy of programming. Creative programmers writing clever code (e.g., MIT Tech Model Railroad Club).

Phase 2 (1970s-mid-1990s):* Negative overtones. Unauthorized break-ins, organized computer theft, and adult white-collar crime.

Phase 3 (mid-1990s-Present):* Growth of Web. Widespread internet access exploited by low-skilled "script kiddies" using pre-written tools to conduct DDoS attacks and fraud.

- **Hacktivism:** Hacking to promote political causes (defacing sites, DDoS). Activists view it as civil disobedience; opponents view it as crime or cyberterrorism because it violates property rights and denies free speech to others.
- **US Laws:**

CFAA (1986/1996):* Illegal to access, alter, damage, or destroy info on a protected computer without authorization.

Patriot Act (2001):* Increased hacking penalties and allowed recovery of server response/recovery costs.

- **Canadian Laws:**

Mischief (s. 430(1)):* Sabotage, hardware destruction, or data alteration. s. 430(1.1) covers mischief to data. s. 430(5.1) covers danger to life.

Colour of Right: *Honest belief that an action is lawful. No conviction if acting with legal justification, excuse, or colour of right. Requires* Mens Rea* (criminal intent).

Viruses:* No specific law banning creation/distribution, but distribution is prosecuted if it causes mischief to data (s. 430(1.1)).

Fraud (s. 380):* Applies to theft/fraud of digital property (e.g., bank credit). s. 321 applies forgery to electronic documents.

Unauthorized Entry:* Prosecuted under s. 380 (fraud), s. 403 (personation), or s. 342.1(1) (dishonest acquisition of computer services).

Hacking Tools: *No general ban in Canada, but s. 327 prohibits tools intended* primarily* to bypass a telecommunication facility (legally distinct from a computer system).

Week 5 Lecture 2: Cyber Crimes

- **Definition:** Subset of computer crime. Any criminal activity involving a computer, networked device, or network.

- **Targets of Cybercrime:**

1. *Individual Property:* Transmitting viruses, unauthorized computer control, intellectual property theft, internet time theft.

2. *Organization:* Possession of unauthorized info, cyberterrorism, software piracy.

3. *Society at Large:* Child pornography, trafficking, financial crimes, online gambling, forgery, sale of illegal articles.

- **Common Cybercrimes:**

Computer Virus:* Program attaching to software to cause damage.

Phishing:* Fake emails containing links to fraudulent sites to steal credentials.

Hacking:* Gaining unauthorized access to data in a computer system or network.

Spoofing:* Pretending to be a trusted entity to gain confidence, access, or steal data.

Netspionage:* Hacking systems to steal confidential data and sell it to third parties.

Cyberstalking:* Online harassment and threatening messages sent over time.

Cyberterrorism:* Cyberattacks against networks/information to coerce or intimidate governments for social/political agendas.

- **Motives:** Entertainment, Profit, Revenge/Infatuation, Political agenda, Sexual, Psychiatric illness.
- **Defenses against Embezzlement/Sabotage:** Rotate employee roles, enforce unique IDs and passwords, implement audit trails, and perform background checks.
- **Identity Theft:** Caused by SSN/SIN abuse, careless handling, and weak database security. Defenses include limiting identifier use, securing databases, credit report monitoring, and consumer education.
- **Forgery:** Enabled by powerful digital manipulation software and high-quality scanners/printers. Defenses include anti-counterfeiting features, verification

methods, and legal/procedural incentives.

- **Crime Fighting vs. Civil Liberties:**

* Automated surveillance vs. warrants based on probable cause.

* Biometric verification vs. database dossier risks.

* Search & seizure of computers vs. business disruption.

* Global Cybercrime Treaty vs. localized privacy standards.

- **Jurisdictional Issues:** Actions legal in one country are prosecuted in others where they are illegal (e.g., Dmitry Sklyarov's arrest in the US for writing software that was legal in Russia).

Part 2: Solved Previous Exam Papers

Paper 1: Mid Semester Examination, Spring-2026 (CSE326)

Q1: Explain how AI, IoT, and automation can enhance learning, campus safety, and student convenience in a smart university? Discuss two key benefits of computer technology in education, security, and communication within a smart university. [5 marks, CO1]

- **Learning (Education):** AI can provide personalized tutoring platforms that adapt to a student's pace, while interactive simulations let students practice complex procedures (e.g., virtual medical surgery or engineering designs) safely.
- **Campus Safety (Security):** IoT sensors can monitor buildings for environmental hazards (fire, gas, moisture), and security cameras combined with facial recognition can track unauthorized access to campus facilities.
- **Student Convenience (Automation):** Smart cards and barcodes simplify library checkouts, and automated billing systems streamline cafeteria payments.
- **Two Key Benefits in Education:**

1. *Simulations and Distance Learning:* Expands access to high-quality interactive education regardless of geography, saving physical infrastructure costs.

2. *Personalized Learning:* Tailors curriculum material to individual student gaps, reducing failure rates.

- **Two Key Benefits in Security:**

1. *Biometrics & Smart Card Access:* Restricts entry to secure dorms and labs to authorized personnel, preventing theft.

2. *Environmental Monitoring:* IoT sensors track acidity, emissions, and temperatures to protect research materials and student health.

- **Two Key Benefits in Communication:**

1. *Non-invasive Broadcasts:* Digital portals and emails allow students to receive notices

at their convenience without interrupting class time.

2. *Immediate Remote Access:* Allows instant retrieval of grades, lecture notes, and library catalog files from home.

Q2(a): Based on the events at the Election Commission, identify and explain specific privacy threats that occurred. [5 marks, CO1]

Based on the slides, three major categories of privacy threats occurred:

- **Unauthorized use or release by "insiders":** The mid-level IT officer at the EC abused their authorized credentials to export and leak private mobile numbers and NID details of 500,000 citizens to a political party's digital wing.
- **Theft of information:** Russian hackers broke into the system and stole the biometric data of deceased and inactive citizens through an unpatched legacy portal.
- **Inadvertent leakage/Negligence:** The EC failed to archive or remove inactive records, maintaining highly sensitive biometric data on a live, unpatched server. This created an unnecessary vulnerability.
- **Secondary Use:** The citizen NID details and mobile numbers collected solely for voter registration were diverted to a political party for "targeted voter outreach" without the citizens' informed consent.

Q2(b): Failure to remove or archive the biometric data of inactive citizens after the 10-year threshold violates which principle of data collection and use? [5 marks]

It violates the **Data Retention** principle under the **Fair Information Principles (FIP)**.

- *FIP Rule:* "Keep data only as long as needed."
- *Justification:* Storing highly sensitive biometric data of deceased citizens or expatriates who have not voted for over 15 years violates this principle. Retaining unneeded data indefinitely increases the vulnerability of data and creates a severe security risk, which in this case was exploited by hackers.

Q3: Explain how different levels of identity identification among attackers influence accountability and ethical responsibility in cyber activities. [5 marks, CO2]

Based on the slides, different levels of pseudonymity alter accountability and ethics:

- **Public Pseudonyms (Traceable to real individuals):**

Accountability:* High. Because the public can link the online identity to the actual physical person, these users are legally vulnerable.

Ethical/Legal Responsibility:* Attackers face direct legal consequences (e.g., prosecution under mischief/sabotage laws). They cannot escape the moral consequences of their actions, aligning with traditional civil disobedience where one accepts the punishment.

- **Non-public Pseudonyms (Known only to platform administrators):**

Accountability:* Moderate. The attacker's identity is shielded from the public but held by the platform. Accountability depends on whether administrators comply with law enforcement warrants to reveal the user's details.

Ethical/Legal Responsibility:* The user operates with a layer of insulation, assuming the platform will protect them. Ethically, they run the risk of exposure if the platform is hacked, audited, or subpoenaed.

- **Unlinkable Pseudonyms / True Anonymity (Cannot be linked to any real person):**

Accountability:* Zero. Since no one holds the mapping, the attacker cannot be traced.

Ethical/Legal Responsibility:* While this level protects whistleblowers, in the context of cyberattacks (like DDoS), it completely removes accountability. It enables anti-social and destructive behavior without any legal consequences, violating property rights and disrupting public services with zero personal risk.

Q4: Analyze the scenario and determine the type of cybercrime, potential targets, and the motives behind the attack, applying principles of computing ethics and security. [5 marks, CO2]

- **Type of Cybercrime:**

1. *Hacking:* Gaining unauthorized access to the e-commerce servers.

2. *Identity Theft / Personation:* Creating fake accounts and assuming customers' identities (violating Personation s.403) to drain digital wallets.

3. *Computer Sabotage (Mischief to Data):* Altering customer delivery details (violating Mischief to Data s.430(1.1)).

4. *Financial Fraud:* Stealing money from digital wallets (s.380).

- **Potential Targets:**

1. *The Organization (E-commerce Server):* The server infrastructure and database were breached.

2. *The Individuals (Customers):* Personal NID details, account credentials, and digital wallet balances were compromised.

- **Motives:**

Profit:* The scenario states the breach was carried out "mainly to earn quick money and fulfill their desire for financial gain."

- **Computing Ethics and Security Application:**

Security failure:* The company failed to protect customer data security (a violation of FIPs). They lacked encryption and access controls to detect anomalous wallet transfers or shipping address edits.

Ethics: *The attackers showed clear* Mens Rea* (criminal intent) to cause financial harm for personal gain, violating the ethical use of technology and individual property rights.

Midterm Examination, Spring 2025 (CSE498/CSE431)

Q1: Analyze how TechDIU's adoption of AI-powered automation has led to both job displacement and job creation in this scenario. Suggest one strategic initiative the company could adopt to minimize the negative impact of automation while promoting workforce sustainability. [5 marks, CO1]

- **Job Displacement:** Replacing routine, entry-level human customer service tasks with AI-driven chatbots and virtual assistants led directly to the layoff of 30% of the workforce.
- **Job Creation:** The adoption created new, higher-value opportunities:

1. *Up-skilling:* Reassigning the remaining 70% of employees to handle more complex customer queries.

2. *New Tech Roles:* Hiring "AI specialists" to oversee and optimize the automated workflows.

3. *New Fields:* Training employees in "AI system management, data analysis, and cybersecurity" creates new classes of jobs.

- **Strategic Initiative:**

Continuous In-house Upskilling and Transition Assistance: *The company should establish a "Redeployment and Retraining Fund" where a portion of the cost savings from AI is used to retrain displaced workers* before* laying them off, qualifying them for the newly created roles (like AI system management or data analysis), ensuring long-term employment.

Q2(a): Explain two major challenges governments and technology companies face when regulating offensive speech and online anonymity while still protecting freedom of expression. Illustrate with real-world examples. [5 marks, CO1]

- **Defining "Offensive" without Censoring Free flow of Ideas:**

Challenge:* Offensive speech is highly subjective. Broad regulations designed to restrict harmful speech can easily be abused by governments or platforms to suppress unpopular political or religious opinions.

Example: *In Canada, the bookstore* Little Sisters* was systematically targeted by customs officials who confiscated gay and lesbian books under the guise of censoring "obscenity."

- **Balancing Criminal Prosecution with Whistleblower Protection:**

Challenge:* Removing online anonymity makes it easy to track down scammers and harassers, but it strips away the safety net for whistleblowers, domestic abuse victims, and political dissidents.

Example:* Activists in countries with restrictive regimes (like China) rely on anonymous browsers (like Tor) and pseudonyms to report government abuse. Enforcing real-name policies exposes them to immediate retribution.

Q2(b): Analyze how this scenario [anonymous blogger exposing corporate corruption] relates to the principles of freedom of speech. Should the blogger have unrestricted freedom to post anonymously, or should there be limitations? Justify your answer. [5 marks]

- **Relation to Free Speech:** The blogger acts as a whistleblower, promoting the *marketplace of ideas* and the *search for truth* by exposing corporate corruption. However, the corporation has a right to protect its reputation against false, damaging statements (libel), which is a key legal limitation to free expression.
- **Limitations and Justification:**

There must be limitations:* Unrestricted anonymity is dangerous because it allows individuals to publish completely fabricated, malicious lies that destroy reputations and cause massive economic damage without any accountability.

Balancing solution: *The blogger should not have* unrestricted *anonymity, but the corporate demand to reveal their identity must meet a very high legal standard. A court should only order the platform to reveal the blogger's identity (non-public pseudonym) if the corporation can present a strong, evidence-backed* prima facie* case showing that the blogger's claims are false, defamatory, and caused direct financial harm. This protects legitimate whistleblowers while preventing anonymous libel.

Q3(a): Explain how modern surveillance technologies impact personal privacy. [5 marks, CO1]

- **Dossier Creation:** GPS, smart cards, and cell phones collect massive amounts of location and transaction details. When combined, these small items provide a detailed profile of a person's private life.
- **Invisible Information Gathering:** Surveillance is conducted from a distance without the individual's knowledge, meaning users have no direct control over their data.
- **Shift in Presumption of Innocence:** Massive databases treat all citizens as suspects, using algorithms to profile and characterize likely criminal behaviors, which results in innocent people being subjected to investigations.
- **Vulnerability of Data:** Keeping sensitive location and communication logs on external servers presents a permanent risk of data leaks and hacking.

Q3(b): A mobile app offers free services but collects detailed user information, including browsing history and location, without notifying users. This data is then sold to advertisers. Examine the opt-in and opt-out policies and its effects in user privacy. [5 marks]

- **App's Actions:** The app is performing **invisible information gathering** (tracking without notice) and **secondary use** (selling data to advertisers), violating the Fair Information Principles.
- **Opt-in Policy & Privacy Effects:**

Mechanism:* The app cannot collect any location or browsing data unless the user takes an affirmative action (e.g., checking an unchecked box) to consent.

Privacy Effect:* High. It protects privacy by default. Users are fully aware of what data is collected and can opt to deny access, stopping secondary uses.

- **Opt-out Policy & Privacy Effects:**

Mechanism:* The app tracks browsing and location by default. The user must manually navigate settings to untick consent options.

Privacy Effect:* Low. Because settings are complex and policies are long, most users are unaware of the tracking. The default state is surveillance, leaving users vulnerable to covert profiling and marketing.

Midterm Examination, Spring 2023 (CSE498)

Q1(a): Do you think agriculture automation is the best solution for the problem in cropping? Give five arguments for your opinion. [5 marks, CO1]

Yes, agriculture automation is the best solution to address modern farming challenges. Five arguments support this:

- **Solves Labor Shortages:** Directly addresses the decline in agricultural labor caused by urbanization, low wages, and aging populations.
- **Increases Efficiency:** Automated machinery can work longer hours and operate under harsh conditions without fatigue.
- **Improves Crop Yields:** Automated precision sensors map crop health and soil quality, optimizing fertilizer and water distribution.
- **Ensures Workplace Safety:** Replaces humans in dangerous, physically demanding, or toxic tasks (e.g., pesticide spraying, heavy lifting).
- **Reduces Waste:** Automated steering and GPS systems prevent overlapping passes during planting and spraying, saving fuel and materials.

Q1(b): Mention some uses of computing in agriculture sectors for high yielding with cost effective plan. [5 marks]

- **Soil & Moisture Sensors (IoT):** Real-time monitoring of soil temperature, moisture, and acidity, allowing automated, targeted watering to conserve resources.

- **Drones & Satellite Analysis:** Capturing multi-spectral crop images to identify pest outbreaks early, limiting pesticide use to infected areas.
- **GPS Autosteer Tractors:** Automating field planting and harvesting pathing to minimize overlaps, reducing fuel usage and planting time.
- **Automated Drip Irrigation:** Scheduling irrigation based on local weather reports and humidity data, reducing water waste.
- **Data Mining & Predictive Modeling:** Analyzing historical crop yields and weather data to forecast production and plan market sales, preventing loss.

Q2(a): Does freedom of speech apply to everything? Explain your thoughts. [5 marks, CO2]

No, freedom of speech does not apply to everything. For a society to remain safe, expression must be subject to legal and ethical limits. Non-protected areas include:

- **Incitement of Violence & Genocide:** Speech that directly encourages physical harm or genocide against an identifiable group is criminal (s. 318 & 319 of Canada's Criminal Code).
- **Obscenity & Child Pornography:** Materials depicting child sexual abuse or extreme, violent sex acts are strictly illegal.
- **Libel and Slander:** Intentionally publishing false, damaging statements about a person or business to harm their reputation (defamation) is legally actionable.
- **Fraud & Deceptive Advertising:** Intentionally lying to consumers for financial gain (e.g., selling fake medicines online) is a crime.
- **National Security Threats:** Exposing sensitive military secrets or causing public panic (e.g., shouting "bomb" in an airport).

Q2(b): What is freedom of expression in internet? Give some examples on it. [5 marks]

- **Definition:** The right of individuals to use digital tools (social media, blogs, emails, websites) to create, publish, access, and share opinions, information, and ideas without arbitrary government censorship.

- **Examples:**

1. *Whistleblowing:* Publishing anonymous posts or documents exposing corporate corruption.

2. *Political Blogging:* Writing articles criticizing a government policy on a personal website.

3. *Online Forums:* Participating in open debates on religious, political, or social networks.

4. *Creative Expression:* Sharing music, videos, and digital art on platforms like YouTube or DeviantArt.

Q2(c): What are some effective strategies individuals can use to protect their privacy online in an era of ubiquitous data collection and surveillance? [5 marks]

- **Employ Encryption:** Use end-to-end encrypted messaging services (like Signal) and VPNs to encrypt data in transit.
- **Use Privacy-Enhancing Technologies (PETs):** Install cookie disablers, ad-blockers, and anti-spyware software on web browsers.
- **Anonymize Searches:** Use privacy-centric search engines (like DuckDuckGo) and browsers (like Tor) to prevent IP tracking.
- **Practice Active Consent:** Read privacy settings on apps, opt-in only when necessary, and untick pre-checked boxes that allow data sharing.
- **Minimize Digital Footprint:** Avoid sharing sensitive location data or personal identifiers (like NID or phone numbers) on public websites.

Midterm Exam - Previous Paper A (Abdullah)

Q1: Explain the benefits of participating in online communities. Provide examples of how individuals can gain knowledge and support through these interactions. [5 marks, CO1]

- **Benefits:**

1. *Intellectual and Collaborative Growth:* Online spaces bring together experts and learners from around the world, allowing for 24/7 collaborative learning.

2. *Social & Emotional Support:* Allows individuals facing rare illnesses, personal crises, or niche hobbies to find peer support, reducing isolation.

3. *Overcoming Physical Boundaries:* Breaks down traditional geographic limits, allowing immediate access to global pools of information.

- **Examples:**

1. *Stack Overflow:* Software developers post programming errors and collaborate on coding solutions, speeding up global learning.

2. *Medical Support Groups:* Patients with rare chronic illnesses share treatment insights and coping mechanisms, gaining emotional strength.

3. *Academic Forums (e.g., ResearchGate):* Researchers share findings and peer-review papers across borders.

Q2: Identify the category of privacy threat illustrated in this scenario [Bank employee mistakenly emails database spreadsheet to client]. Explain your answer with justification, describing why it fits this category and not the others. [5 marks, CO2]

- **Category: Inadvertent leakage of information**.
- **Justification:** The employee *mistakenly* attached the customer spreadsheet and emailed it to an external client. There was no intent (*Mens Rea*) to compromise security, nor was it a deliberate release. It was a simple human

error, which directly defines "inadvertent leakage."

- **Why it does not fit the others:**

Unauthorized use/release by insiders: *This requires the employee to have* deliberately* shared or sold the data without authorization. Here, it was an accident.

Theft of information:* This refers to external hackers breaking into the bank's servers to extract databases. No security breach occurred.

Intentional, institutional uses:* This refers to the bank purposely using customer records for secondary purposes like profiling or target marketing.

Q3: Identify and explain the Fair Information Principles that Arafat should follow to collect, store, and use the data responsibly [Health survey, cloud storage, large data collection]. [5 marks, CO2]

- **Informed Consent (Notice and Consent):** Arafat must clearly inform the students before they fill out the survey about what health data is collected, how it will be stored, and how it will be used, obtaining their explicit consent.
- **Collection Limitation:** Arafat must limit data collection strictly to what is necessary for the public health study, rather than trying to "gather as much data as possible."
- **Data Retention:** He must establish a policy to delete or anonymize the data once the research paper is completed, rather than keeping it indefinitely.
- **Security Safeguards:** Storing sensitive health and habit records on a personal cloud account is a risk. He must protect the data using encryption, strong passwords, and preferably host it on secure university servers.
- **Use Limitation (Secondary Use):** The data must only be used for the stated public health research and cannot be shared with third parties without consent.

Q4: Evaluate how online platforms can balance the right to free speech with the need to prevent offensive or harmful content. What ethical and social considerations should guide their decisions? [5 marks, CO1]

- **Balancing Strategy:**

1. *Clear Terms of Service:* Establish transparent, objective definitions of hate speech and harassment.

2. *Hybrid Moderation:* Use AI algorithms to scan and flag content, but rely on human reviews to evaluate context and prevent over-blocking.

3. *Tiered Enforcement:* Use warnings, fact-check labels, or demote content algorithmically (reducing reach) rather than resorting directly to account deletion.

- **Ethical and Social Considerations:**

1. *Harm vs. Censorship:* Platforms must protect marginalized groups from harassment while avoiding the censorship of controversial political views, keeping the *marketplace of ideas* open.

2. *Algorithmic Responsibility:* Ensuring recommendation algorithms do not actively promote extreme, controversial content just to boost screen-time and ad revenue.

3. *Jurisdictional Compliance:* Respecting local standards (like Canada's hate speech laws) while preserving general free speech values.

Q5: Based on the scenario [Bank email scam stealing credentials], identify the type of cybersecurity attack involved and discuss the preventive measures organizations and consumers can take to protect themselves against such attacks. [5 marks, CO1]

- **Type of Attack: Phishing** and **Spoofing**.

Justification:* The attacker sent fake emails appearing to come from a legitimate bank (spoofing) containing links to harvest login details (phishing) for financial theft.

- **Preventive Measures for Organizations:**

1. *Implement Authentication Protocols:* Enforce SPF, DKIM, DMARC to block spoofed emails from reaching users.

2. *Require Multi-Factor Authentication (MFA):* Enforce MFA so stolen credentials alone cannot grant access to accounts.

3. *Customer Education:* Explicitly notify customers that the bank will never ask for personal details or passwords via email links.

- **Preventive Measures for Consumers:**

1. *Examine the Sender Address:* Closely check for subtle misspellings in the sender's domain.

2. *Avoid Clicking Email Links:* Go directly to the official banking site by typing the URL in the browser.

3. *Never share credentials:* Keep login details secure, and use updated browser security extensions.

Good luck with your exam tomorrow! This guide covers all slides and previous papers in detail. Retain these terms (FIPs, Mischief s.430, Opt-in/Opt-out, Anonymity levels) to score full marks!